

Discrete distributions for the claim frequency

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Discrete distributions for the claim frequency

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Bernoulli distribution

Definition

We have a random experiment with two possible outcomes: success or failure. The probability of success is p and the probability of failure is $1 - p = q$ with $0 \leq p \leq 1$. Then, X is **Bernoulli**, $X \sim Be(p)$.

- **Probability function**

$$P(X = 1) = p, \quad P(X = 0) = 1 - p = q$$

$$P(X = x) = p^x(1 - p)^{1-x}, x = 0, 1.$$

- **Mean:**

$$E(X) = 1 \cdot p + 0 \cdot q = p$$

- **Variance:**

$$Var(X) = E[X^2] - E[X]^2 = p - p^2 = p(1 - p) = pq$$

Bernoulli distribution

Moments

- The moment of order k with respect to the origin is

$$E(X^k) = 0^k P(X = 0) + 1^k P(X = 1) = p.$$

- In general, the moment generating function is

$$M_{X_{Be}}(t) = E[e^{tX}] = 1 - p + pe^t$$

Binomial

Definition

We do n Bernoulli experiments in the following conditions:

- the experiments are identical, that is to say, the probability of success p is the same for all of them.
- the experiments are independent, that is to say, the result of one experiment does not have any effect on the others.

Binomial distribution: we do n independent Bernoulli experiments, with probabilities of success equal to p . The random variable $X = \text{Number of successes in the } n \text{ experiments}$, is called binomial, and the corresponding distribution, is called Binomial distribution. It is represented as $X \sim B(n, p)$. The probability function is:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \text{ si } x = 0, 1, 2, \dots, n.$$

Binomial distribution

Definition

- Therefore, we can represent the Binomial distribution as the sum of Bernoulli variables. If $z_1, z_2, \dots, z_n$ are independent and identically $Be(p)$ distributed random variables then the variable $X = z_1 + z_2 + \dots + z_n$ has a Binomial distribution, so that $X = z_1 + z_2 + \dots + z_n \sim B(n, p)$.
- **Moment generating function:**

$$\begin{aligned}
 M_{X_B}(t) &= E[e^{tX}] = E[e^{t(z_1+z_2+\dots+z_n)}] = E[e^{tz_1}]E[e^{tz_2}]\dots E[e^{tz_n}] \\
 &= [M_{X_{Be}}(t)]^n = (1 - p + pe^t)^n
 \end{aligned}$$

as z_i have moment generating function $M_{X_{Be}}(t) = 1 - p + pe^t$.

Binomial distribution

Moments

- Moments:**

$$\begin{aligned}
 E(X) &= np \\
 Var(X) &= np(1-p) \\
 \gamma_1 &= \frac{1-2p}{\sqrt{np(1-p)}} \\
 \gamma_2 &= \frac{1-6p(1-p)}{np(1-p)}
 \end{aligned}$$

Binomial distribution

Properties

- **Convolutions:** Si $x_1, x_2, \dots, x_m$ are independent random variables such that

$$X_1 \sim B(n_1, p), X_2 \sim B(n_2, p), \dots, X_m \sim B(n_m, p)$$

entonces

$$X_1 + X_2 + \dots + X_m \sim B(n_1 + n_2 + \dots + n_m, p)$$

Proof: by using the moment generating functions

$$\begin{aligned}
 M_{X_1+X_2+\dots+X_m}(t) &= E[e^{t(X_1+X_2+\dots+X_m)}] = E[e^{tX_1}] \dots E[e^{tX_m}] \\
 &= (1 - p + pe^t)^{n_1} \dots (1 - p + pe^t)^{n_m} \\
 &= (1 - p + pe^t)^{n_1+n_2+\dots+n_m}
 \end{aligned}$$

which is the moment generating function of $B(n_1 + n_2 + \dots + n_m, p)$.

Poisson distribution

Definition

The **Poisson distribution** is used as a probability model for the occurrence of rare or unfrequent events. Moreover, it is a limiting case of the Binomial distribution when the number of experiments is big and the probability of success is low. We assume that X represents the number of occurrences per time or space unit, for example:

- number of claims in a year in an insurance company
- number of traffic accidents in a week in a specific part of a road

Poisson distribution

Definition

To get the probability function of X we establish the following hypothesis:

- the probability of occurrence of a rare event in a small time interval d is proportional to its duration, that is to say, $p = cd$ where c is a positive constant.
- The probability of occurrence of two or more events in the same interval is despreciable.
- The number of events in an interval is independent of the number of events in other disjunts intervals.
- The number of events that take place in two intervals of the same duration have the same probability distribution.

Poisson distribution

Definition

We divide the interval t in n subintervals of duration $d = t/n$. By considering the hypothesis that we did, we can think of the n subintervals as n independent and identical Bernoulli experiments $X_1, X_2, \dots, X_n$ with

$P(X_i = 1) = p, i = 1, \dots, n$.

Now, $X = \sum_{i=1}^n X_i$ follows a Binomial distribution with parameters n and $p = ct/n$. We want to obtain the probability function of X when $n \rightarrow \infty$. It can be proved that

$$\lim_{n \rightarrow \infty} \binom{n}{x} p^x (1-p)^{n-x} = \frac{e^{-ct} (ct)^x}{x!}, x = 0, 1, \dots$$

if we consider $\lambda = ct$ then we get:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, \dots; \lambda > 0.$$

Poisson distribution

Definition

The limit of the Binomial distribution is the Poisson distribution, when $n \rightarrow \infty$, $p \rightarrow 0$ and the product $\lambda = np$ remains constant. Proof: as $\lambda = np$ constant, then $p = \lambda/n$, and we substitute and calculate the limit when $n \rightarrow \infty$,

$$\begin{aligned}
 \lim_{n \rightarrow \infty} P(X = x) &= \lim_{n \rightarrow \infty} \binom{n}{x} p^x (1-p)^{n-x} = \\
 &= \lim_{n \rightarrow \infty} \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
 &= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \frac{n(n-1)\dots(n-x+1)}{n^x} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x} \\
 &= \frac{\lambda^x}{x!} \cdot 1 \cdot e^{-\lambda} \cdot 1 = \frac{e^{-\lambda} \lambda^x}{x!}
 \end{aligned}$$

Poisson distribution

Definition

Poisson distribution: The random variable X is Poisson distributed and it is represented as $X \sim P(\lambda)$ if its probability law is as follows

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, \dots; \lambda > 0.$$

- **Moment generating function:**

$$\begin{aligned} M_{X_P}(t) &= E[e^{tX}] = \sum_{n=0}^{\infty} e^{tn} \frac{e^{-\lambda} \lambda^n}{n!} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda e^t)^n}{n!} = e^{\lambda(e^t - 1)} \end{aligned}$$

Poisson distribution

Definition

- We therefore have that:

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

$$\gamma_1 = \frac{1}{\sqrt{\lambda}}$$

$$\gamma_2 = \frac{1}{\lambda}$$

Poisson distribution

Properties

- **Convolutions:** Let $X_i \sim P(\lambda_i), i = 1, 2, \dots, n$ be independent Poisson random variables. The, the random variable sum $X_1 + X_2 + \dots + X_n$ is also Poisson

$$X_1 + X_2 + \dots + X_n \sim P(\lambda_1 + \lambda_2 + \dots + \lambda_n)$$

Proof: by using the moment generating functions we have:

$$\begin{aligned}
 M_{X_1+X_2+\dots+X_n}(t) &= E[e^{t(X_1+X_2+\dots+X_n)}] \\
 &= E[e^{tX_1}]E[e^{tX_2}]\dots E[e^{tX_n}] \\
 &= e^{\lambda_1(e^t-1)}e^{\lambda_2(e^t-1)}\dots e^{\lambda_n(e^t-1)} \\
 &= e^{(\lambda_1+\lambda_2+\dots+\lambda_n)(e^t-1)}
 \end{aligned}$$

and therefore $X_1 + X_2 + \dots + X_n \sim P(\lambda_1 + \lambda_2 + \dots + \lambda_n)$.

Poisson distribution

Estimation

As we know, the probability function of a Poisson distributed variable is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, \dots; \lambda > 0.$$

Therefore, for a random sample of size n , $X_1, X_2, \dots, X_n$, we have that the likelihood function is

$$L(\lambda; X) = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n X_i}}{\prod_{i=1}^n X_i!}$$

where the statistic $\sum_{i=1}^n X_i$ is sufficient, complete and unbiased for the estimation of λ . It can be proved that the ML estimator of λ is the sample average

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i$$

Geometric distribution

Definition

We do independent Bernoulli experiments with probability of success equal to p . We define the random variable X = Number of failures before the first success. The probability law is:

$$P(X = k) = (1 - p)^k p, k = 0, 1, 2, \dots$$

where $0 < p < 1$, and we represent $X \sim Ge(p)$. A second parametrization is

$$P(X = k) = \frac{1}{1 + \beta} \left(\frac{\beta}{1 + \beta} \right)^k, k = 0, 1, 2, \dots$$

where $\beta > 0$. In this case $X \sim Ge(p = 1/(1 + \beta))$.

Geometric distribution

Introduction

- **Moment generating function:**

$$M_{X_{\text{Ge}}}(t) = E[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} p(1-p)^k = \frac{p}{1 - (1-p)e^t}$$

if $|e^t| < 1/(1-p)$, or according to the second parametrization:

$$M_{X_{\text{Ge}}}(t) = E[e^{tX}] = \frac{1}{1 - \beta(e^t - 1)}$$

si $|e^t| < (1 + \beta)/\beta$.

Geometric distribution

Moments

- Moments:**

$$E(X) = \frac{1-p}{p} = \beta$$

$$Var(X) = \frac{1-p}{p^2} = \beta(1 + \beta)$$

- Other properties:** the geometric distribution satisfies the memoryless property, if $X \sim Ge(p)$, $P(X \geq n + m | X \geq m) = P(X \geq n)$ for $n, m = 0, 1, 2, \dots$

Negative Binomial distribution

Definition

The **Negative binomial** distribution is a generalization of the geometric distribution. We assume that we carry out independent Bernoulli experiments with probability of success p . We define the random variable: $X = \{\text{Number of failures before the } r\text{-success}\}$. The probability law of this variable, which is negative binomial, is:

$$P(X = k) = \binom{k+r-1}{k} p^r (1-p)^k, k = 0, 1, 2, \dots$$

where $0 < p < 1$, $r > 0$, and we represent $X \sim BN(r, p)$. It is clear that the geometric distribution is a particular case of the Negative Binomial distribution with $r = 1$.

Negative Binomial distribution

Moments

A second parametrization consist on defining $p = 1/(1 + \beta)$, therefore

$$P(X = k) = \binom{k + r - 1}{k} \left(\frac{1}{1 + \beta} \right)^r \left(\frac{\beta}{1 + \beta} \right)^k, k = 0, 1, 2, ..$$

where $\beta > 0$ and $r > 0$. In this case, we say that $X \sim BN(r, p = 1/(1 + \beta))$.

- **Moment generating function:**

$$M_{X_{BN}}(t) = E[e^{tX}] = \frac{p^r}{[1 - (1 - p)e^t]^r}, \text{ if } |e^t| < 1/(1 - p).$$

or, according to the second parametrization

$$M_{X_{BN}}(t) = E[e^{tX}] = \frac{1}{[1 - \beta(e^t - 1)]^r}, \text{ if } |e^t| < (1 + \beta)/\beta.$$

Negative Binomial distribution

Moments

- Moments:**

$$E(X) = \frac{r(1-p)}{p} = r\beta$$

$$Var(X) = \frac{r(1-p)}{p^2} = r\beta(1+\beta)$$

$$\gamma_1 = \frac{2-p}{\sqrt{r(1-p)}}$$

Therefore, this distribution is overdispersed.

Negative Binomial distribution

Moments

- **Convolutions:** if $X_1 \sim BN(r_1, p), \dots, X_n \sim BN(r_n, p)$ are independent random variables, then $X_1 + X_2 + \dots + X_n$ is also Negative Binomial with parameters $r_1 + r_2 + \dots + r_n$ and p .

$$\begin{aligned}
 M_{X_1+X_2+\dots+X_n}(t) &= E[e^{t(X_1+X_2+\dots+X_n)}] \\
 &= E[e^{tX_1}]E[e^{tX_2}]\dots E[e^{tX_n}] \\
 &= \frac{p^{r_1}}{[1 - (1-p)e^t]^{r_1}} \cdots \frac{p^{r_n}}{[1 - (1-p)e^t]^{r_n}} \\
 &= \frac{p^{r_1+\dots+r_n}}{[1 - (1-p)e^t]^{r_1+\dots+r_n}}
 \end{aligned}$$

Negative Binomial distribution

Moments

The **Negative Binomial** distribution is also known as **mixt Gamma-Poisson distribution**, because the distributionn Negative Binomial, $BN(r, p)$ is a Poisson distribution $P(\lambda)$ where λ is a random variable, with a $Gamma(r, p/(1 - p))$ distribution.

Proof: A random variable is $X \sim Gamma(\alpha, \sigma)$ if it has the following density function

$$f(x; \alpha, \sigma) = \frac{1}{\sigma^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\sigma}, x > 0$$

where $\alpha, \sigma > 0$. Therefore, we have a $Gamma(r, p/(1 - p))$ if it has the following density function

$$f(x; \alpha, \sigma) = \frac{1}{\left(\frac{p}{1-p}\right)^r \Gamma(r)} x^{r-1} e^{-x(1-p)/p}, x > 0$$

Negative Binomial distribution

Properties

We consider X which has a $BN(r, p)$ distribution. We have to proof that X is a Poisson random variable $P(\lambda)$ where λ is $Gamma(r, p/(1-p))$ distributed:

$$\begin{aligned}
 f(x) &= \int_0^{\infty} f_{Poisson(\lambda)}(x) \cdot f_{Gamma(r, \frac{p}{1-p})}(\lambda) d\lambda \\
 &= \int_0^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} \cdot \frac{\lambda^{r-1} e^{-\lambda(1-p)/p}}{\left(\frac{p}{1-p}\right)^r \Gamma(r)} d\lambda \\
 &= \frac{(1-p)^r p^{-r}}{x! \Gamma(r)} \int_0^{\infty} \lambda^{r+x-1} e^{-\lambda/p} d\lambda \\
 &= \frac{(1-p)^r p^{-r}}{x! \Gamma(r)} p^{r+x} \Gamma(r+x) \\
 &= \frac{\Gamma(r+x)}{\Gamma(x+1) \Gamma(r)} (1-p)^r p^x
 \end{aligned}$$

Negative Binomial distribution

Properties

By using this result, we can proof again that the Negative Binomial is overdispersed, that is to say, the variance is higher than the mean. The mean of N will be

$$\begin{aligned}
 E[N] &= E[E[N|\lambda]] = E[\lambda] \\
 Var[N] &= E[Var[N|\lambda]] + Var[E[N|\lambda]] = E[\lambda] + Var[\lambda] > E[\lambda]
 \end{aligned}$$

Negative Binomial distribution

Properties

We say that the random variable S is a **composed distribution**, if the moment generating function can be written as

$$M_S(t) = M_N[\log M_X(t)]$$

where M_N and M_X are two generating functions. The random variable N with generating function $M_N(\cdot)$ is called the primary distribution while the random variable X with generating function M_X is called the secondary function.

Negative Binomial distribution

Properties

Composed distributions arise naturally as the distribution of the random variable

$$S_N = X_1 + X_2 + \dots + X_N$$

as we assume that $X_1, X_2, \dots$ are identically distributed and moreover $N, X_1, X_2, \dots$ are mutually independent, we have that

$$\begin{aligned}
 M_S(t) &= E[e^{tS}] = E[E[e^{tS} | N]] \\
 &= E[M_X(t)^N] = E[e^{N \log M_X(t)}] \\
 &= M_N[\log M_X(t)]
 \end{aligned}$$

Negative Binomial distribution

Properties

Example: we consider the composed distribution S with primary distribution is Poisson $N \sim P(\lambda)$ and the secondary distribution is logarithmic, $X \sim \text{Log}(\theta)$. Then, the composed distribution of S is a Negative Binomial $BN(r, p)$ where $r = -\lambda / \ln(1 - \theta)$ y $p = 1 - \theta$.

Negative Binomial distribution

Properties

A random variable X has a logarithmic distribution $X \sim \text{Log}(\theta)$ if its probability function is given by

$$P(X = k) = -\frac{1}{\log(1 - \theta)} \frac{\theta^k}{k}, k = 1, 2, \dots \text{ y } 0 < \theta < 1.$$

- Moment generating function:**

$$M_{X_{\text{Log}}}(t) = E[e^{tX}] = \frac{\log(1 - \theta e^t)}{\log(1 - \theta)}$$

$$E(X) = \frac{a\theta}{1 - \theta}$$

$$\text{Var}(X) = \frac{a\theta(1 - a\theta)}{(1 - \theta)^2} = \mu \left(\frac{1}{1 - \theta} - \mu \right)$$

where $a = -1/\log(1 - \theta)$ and $\mu = E(X)$.

Negative Binomial distribution

Properties

Example: we consider a composed distribution S with primary distribution Poisson $N \sim P(\lambda)$ and secondary distribution logarithmic, $X \sim \text{Log}(\theta)$. Then, the composed distribution of S is a Negative Binomial $BN(r, p)$ where $r = -\lambda / \ln(1 - \theta)$ and $p = 1 - \theta$.

$$\begin{aligned}
 M_S(t) &= \exp \{ \lambda [M_X(t) - 1] \} \\
 &= \exp \left\{ \lambda \left[\frac{\ln(1 - \theta e^t)}{\ln(1 - \theta)} - 1 \right] \right\} \\
 &= \exp \left\{ \frac{\lambda}{\ln(1 - \theta)} \ln \left(\frac{1 - \theta e^t}{1 - \theta} \right) \right\} \\
 &= \left[\frac{1 - \theta}{1 - (1 - (1 - \theta))e^t} \right]^{-\lambda / \ln(1 - \theta)}
 \end{aligned}$$

which is the moment generating function of a Negative Binomial distribution with the parameters that we referred before.

Negative Binomial distribution

Estimation

We have a Negative Binomial variable $BN(r, p)$ with probability function

$$P(X = x) = \binom{x+r-1}{k} p^r (1-p)^x, k = 0, 1, 2, \dots$$

if we take into account that

$$\binom{x+r-1}{k} = \frac{(x+r-1)!}{x!(r-1)!} = \frac{\Gamma(x+r)}{\Gamma(r)\Gamma(x+1)}$$

From a random sample of size n , it can be proved that the log of the likelihood function is

$$\begin{aligned} l(p, r) &= \sum_{i=1}^n \log \Gamma(X_i + r) - n \log \Gamma(r) - \sum_{i=1}^n \log \Gamma(X_i + 1) \\ &\quad + nr \log p + \sum_{i=1}^n X_i \log(1 - p) \end{aligned}$$

Negative Binomial distribution

Estimation

If we calculate the partial derivatives of this function with respect to p and r and making them equal to 0, we obtain the likelihood equations,

$$\begin{aligned}\hat{p} &= \frac{\hat{r}}{\bar{X} + \hat{r}} \\ \log \hat{r} &= \frac{\Gamma'(\hat{r})}{\Gamma(\hat{r})} - \frac{1}{n} \sum_{i=1}^n \frac{\Gamma'(X_i + \hat{r})}{\Gamma(X_i + \hat{r})}\end{aligned}$$

We substitute the value of $\hat{p}$ in the second equation, and we obtain an equation on $\hat{r}$

$$\log \hat{r} - \log(\bar{X} + \hat{r}) - \frac{\Gamma'(\hat{r})}{\Gamma(\hat{r})} + \frac{1}{n} \sum_{i=1}^n \frac{\Gamma'(X_i + \hat{r})}{\Gamma(X_i + \hat{r})} = 0$$

and its solution gives us the ML estimator of r , that can be obtained numerically.

The (a,b,0) class

Definition

Let X be a discrete non-negative random variable with probability function $p_k = P(X = k)$, $k = 0, 1, 2, \dots$. We say that X belongs to the $(a, b, 0)$ class if it fulfills the following condition

$$p_k = \left(a + \frac{b}{k} \right) p_{k-1}, \quad k = 1, 2, \dots$$

with $p_0 > 0$. The $(a, b, 0)$ class is determined by the probability of 0, p_0 and by the constants a and b . The Binomial, Poisson, Geometric and Negative Binomial distributions are the only distributions that belong to the $(a, b, 0)$ class.

The (a,b,0) class

Definition

Distribución	a	b	p_0
Binomial	$-\frac{p}{1-p}$	$(n+1)\frac{p}{1-p}$	$(1-p)^n$
Poisson	0	λ	$e^{-\lambda}$
Geometric	$\frac{\beta}{1+\beta}$	0	$\frac{1}{1+\beta}$
Negative Binomial	$\frac{\beta}{1+\beta}$	$(r-1)\frac{\beta}{1+\beta}$	$\frac{1}{(1+\beta)^r}$

The (a,b,0) class

Properties

The (a,b,0) distributions have two advantages principales:

- The probabilities can be easily evaluated recursively
- There exist a recursion formula (the so-called Panjer recursion) that let us to evaluate the aggregated claim distribution, when the number of claims belong to that class and the cost is a random variable that can be continuous or discrete.

Zero inflated distributions

Definition

Let Y be a discrete random variable, with probability function $P(Y)$. We define the **zero inflated** random variable as the variable $X = ZY$ where Z follows a *Bernoulli*($1 - \pi$) distribution. The probability function of X is:

$$P(X = 0) = \pi + (1 - \pi)P(Y = 0), \text{ and}$$

$$P(X = k) = (1 - \pi)P(Y = k) \text{ for } k \geq 1.$$

Zero inflated distributions

Definition

If Y is Poisson distributed, we say that X is a Zero Inflated Poisson (ZIP) and it has probability function:

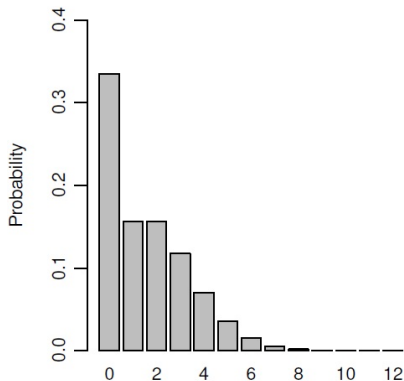
$$P(x) = \begin{cases} \pi + (1 - \pi)\exp(-\lambda) & y = 0 \\ (1 - \pi)\frac{\lambda^y e^{-\lambda}}{y!} & y = 1, 2, \dots \end{cases}$$

It is easy to proof that the expected value of Y is given by $\mu = (1 - \pi)\lambda$ and the variance is given by $\mu + \pi(\mu^2)/(\lambda - \mu)$.

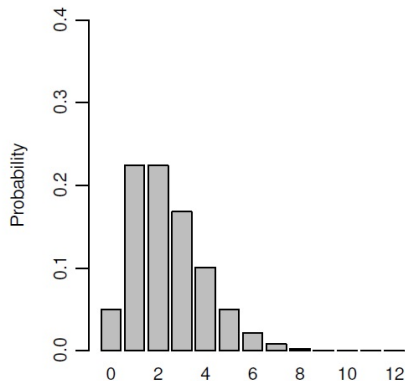
Zero inflated distributions

Example: ZIP

ZIP($\pi = 0.3, \lambda = 3$)



ZIP($\pi = 0, \lambda = 3$) = Poi($\lambda = 3$)



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